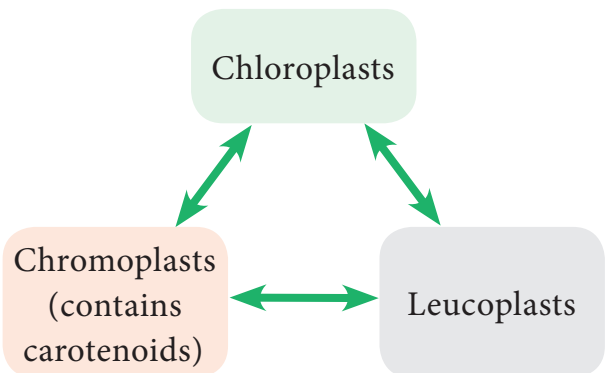
time faster than DNA in the nucleus.

6.6.5 Plastids

The term plastid is derived from the Greek word ***Platikas*** (formed/moulded) and used by **A.F.U. Schimper** in 1885. He classified plastids into following types according to their structure, pigments and function. Plastids multiply by fission.

Plastids	
Chromoplasts	**Leucoplasts**
(Coloured Plastids)	(Colourless Plastids store food materials)
Chloroplast Occurs in green algae and higher plants Pigments chlorophyll *a* and *b*	**Amyloplast** – stores – starch
Phaeoplast Brown algae and dinoflagellates Pigment fucoxanthin	**Elaioplast** – store – lipids (oils) Seed of monocot and dicots.
Rhodoplast Red algae Pigment Phycoerythrin	**Aleuroplast** (or) **Proteoplast** store – Protein

According to Schimper, different kind of plastids can transform into one another.



6.6.6 Chloroplast

Chloroplasts are vital organelle found in green plants. Chloroplast has a double membrane the outer membrane and the inner membrane separated by a space called **periplastidial space**. The space enclosed by the inner membrane of chloroplast is filled with gelatinous matrix, lipo-proteinaceous fluid called **stroma**. Inside the stroma there is flat interconnected sacs called **thylakoid**. The membrane of thylakoid enclose a space called **thylakoid lumen**.

Grana (singular: Granum) are formed when many of these thylakoids are stacked together like pile of coins. Light is absorbed and converted into chemical energy in the granum, which is used in stroma to prepare carbohydrates. Thylakoid contain chlorophyll pigments. The chloroplast contains osmophilic granules, 70s ribosomes, DNA (circular and non histone) and RNA. These chloroplast genome encodes approximately 30 proteins involved in photosynthesis including the components of photosystem I & II, cytochrome bf complex and ATP synthase. One of the subunits of Rubisco is encoded by chloroplast DNA. It is the major protein component of chloroplast stroma, single most abundant protein on earth. The thylakoid contain small, rounded photosynthetic units called **quantosomes**. It is a semi-autonomous organelle and divides by fission (Figure 6.19).

Functions:

- Photosynthesis
- Light reactions takes place in granum,
- Dark reactions take place in stroma,
- Chloroplast is involved in photo-respiration.